

Cognition as Meta-Capital

An Axiomatic Theory of Epistemic Sovereignty, Dual-Triple Intelligence, and Endogenous Sovereign Selection

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Abstract

This paper extends the Six-Nation Bound framework into a general theory of epistemic sovereignty, collective intelligence, and endogenous sovereign formation.

The original binary epistemic structure consisting of Knowledge (K) and Propaganda (P) is generalized into a four-valued epistemic algebra by introducing Silence (S) and Asymmetric Information (A). Intelligence is subsequently introduced as a Boolean state variable generated by informational states. The theory is then extended further by allowing intelligence within one Triple to depend upon informational outputs generated by the opposing Triple.

The resulting structure transforms sovereignty from an exogenous political property into an endogenous outcome generated by a larger cognitive system. Cognition emerges as a recursive form of meta-capital whose outputs include knowledge, institutions, technologies, warfare capabilities, and ultimately sovereign selection itself.

The paper develops logical, economic, financial, geopolitical, military, statistical, and machine-learning interpretations of this framework.

The paper ends with “The End”

1 Introduction

Classical political economy treats sovereignty as a primitive.

States exist. Governments rule. Power is exercised.

Modern information systems challenge this assumption.

Artificial intelligence, machine learning, large-scale computation, and global information networks suggest that sovereignty may be generated by informational processes rather than merely inherited from territorial arrangements.

This paper investigates the consequences of treating sovereignty as an emergent outcome of cognition.

2 The Original Six-Nation Framework

2.1 Triples

Definition 1. *A Triple is an ordered collection of three epistemically sovereign nations*

$$T = \{n_1, n_2, n_3\}.$$

Axiom 1 (Existence). *At least one Triple exists.*

Axiom 2 (Epistemic Induction). *The existence of one Triple induces the existence of a second Triple.*

Denote these by

$$T_1, \quad T_2.$$

2.2 Binary Informational Outputs

Originally,

$$\Omega_0 = \{K, P\}$$

where

- K denotes knowledge,
- P denotes propaganda.

The informational mapping is

$$\mathcal{I} : T_1 \cup T_2 \rightarrow \Omega_0.$$

3 Four-Valued Epistemic Algebra

The binary informational ontology is expanded.

Definition 2. *The generalized informational state space is*

$$\Omega = \{S, K, P, A\}$$

where

$$\begin{aligned} S &= \text{Silence}, \\ K &= \text{Knowledge}, \\ P &= \text{Propaganda}, \\ A &= \text{Asymmetric Information}. \end{aligned}$$

A convenient representation is

$$\Omega = \{0, 1\}^2.$$

Specifically,

$$S = (0, 0),$$

$$K = (1, 0),$$

$$P = (0, 1),$$

$$A = (1, 1).$$

3.1 Interpretation

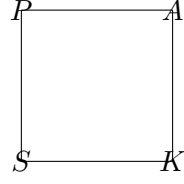
Silence represents the absence of both knowledge and propaganda.

Asymmetric information represents simultaneous presence of both.

State	Representation
S	$(0, 0)$
K	$(1, 0)$
P	$(0, 1)$
A	$(1, 1)$

4 Epistemic Geometry

The four-state algebra forms a Boolean square.



The original framework occupied only the vertices K and P .
The generalized framework occupies the entire square.

5 Intelligence as an Endogenous Variable

Definition 3. *For each Triple define a Boolean intelligence state*

$$I_i \in \{0, 1\}.$$

Initially,

$$I_i = f(S_i, K_i, P_i, A_i).$$

Intelligence is generated internally from informational states.

5.1 Minimal Construction

A minimal specification is

$$I_i = A_i.$$

In this interpretation intelligence corresponds to the capacity to simultaneously process contradictory informational contents.

5.2 General Construction

More generally,

$$I_i = f(S_i, K_i, P_i, A_i).$$

No restriction is imposed upon f beyond logical consistency.

6 Cross-Triple Intelligence Coupling

The central extension of this paper is the introduction of mutual dependence.

$$I_1 = f(S_1, K_1, P_1, A_1, S_2, K_2, P_2, A_2),$$

$$I_2 = f(S_1, K_1, P_1, A_1, S_2, K_2, P_2, A_2).$$

Intelligence ceases to be local.
Instead it becomes relational.

Proposition 1. *If*

$$\frac{\partial I_1}{\partial \Omega_2} \neq 0$$

and

$$\frac{\partial I_2}{\partial \Omega_1} \neq 0,$$

then intelligence is jointly produced.

Proof. The intelligence of each Triple depends upon informational states generated by the other Triple. Therefore neither Triple can independently generate its complete intelligence state. Hence intelligence is co-produced. □

7 The Adversarial Cognition Principle

Traditional geopolitical theories treat rivals as competitors.

In the present framework rivals become cognitive inputs.

Theorem 1 (Adversarial Cognition Principle). *Suppose*

$$\frac{\partial I_1}{\partial \Omega_2} > 0, \quad \frac{\partial I_2}{\partial \Omega_1} > 0.$$

Then elimination of either Triple weakens the intelligence-generating capacity of the overall system.

Proof. If one Triple is removed, informational inputs from that Triple become identically zero.

By hypothesis intelligence is increasing in those inputs.

Therefore aggregate intelligence declines. □

8 Endogenous Sovereign Selection

The original framework treated sovereignty as exogenous.

We now reverse the causal ordering.

Definition 4. *Let*

$$\Sigma$$

denote a sovereign-selection operator.

$$\Sigma : (\Omega_1, \Omega_2, I_1, I_2) \rightarrow Q.$$

The selected sovereign is

$$Q.$$

9 Sovereignty as an Emergent Variable

The original ordering was

$$\text{Sovereignty} \rightarrow \text{Information}.$$

The new ordering becomes

$$\text{Information} \rightarrow \text{Intelligence} \rightarrow \text{Sovereignty}.$$

Thus sovereignty becomes an output.

Theorem 2. *If intelligence is jointly produced and sovereignty is selected by Σ , then sovereignty is an emergent property of the dual-Triple system.*

Proof. The sovereign is generated from informational and intelligence states.

Those states are jointly generated by the coupled system.

Hence sovereignty is generated by the coupled system. □

10 Toward a Theory of Cognition

The previous sections imply a deeper conclusion.

Knowledge is not the ultimate scarce resource.

Neither is information.

Instead the scarce resource becomes cognition itself.

Define

$$C_t$$

to be aggregate cognition.

Then

$$C_t = g(\Omega_t, I_t).$$

The resulting system evolves according to

$$\Omega_t \rightarrow I_t \rightarrow C_t \rightarrow \Omega_{t+1}.$$

The process becomes recursive.

11 Economics of Cognition

11.1 From Information to Meta-Capital

Classical economics recognizes physical capital and labor as primary factors of production.

Human-capital theory subsequently introduced knowledge and skills as productive assets.

The present framework introduces a higher-order factor:

$$C = \text{Cognition.}$$

Unlike ordinary capital, cognition modifies the production process itself.

$$Y = F(K, L)$$

becomes

$$Y = F(K, L; C).$$

More generally,

$$F = F_C.$$

Cognition alters the production function rather than merely entering as an argument.

Definition 5. *Meta-capital is any productive asset capable of modifying the productive efficiency of other productive assets.*

Proposition 2. *Cognition constitutes meta-capital.*

Proof. Cognition generates discoveries, algorithms, institutions, and organizational innovations.

These alter the productivity of labor and capital.

Therefore cognition modifies the production function itself. □

11.2 Recursive Production

The distinctive feature of cognition is self-production.

$$C_{t+1} = H(C_t, \Omega_t, I_t).$$

Thus cognition behaves unlike conventional capital.

Machines do not typically produce machines.

Cognition regularly produces cognition.

12 The Economics of Sovereignty

The original Six-Nation Bound implicitly assumes

$$\text{Power} = \text{Information Production.}$$

The extended framework suggests

$$\text{Power} = \text{Cognition Production.}$$

Consequently geopolitical competition shifts from territorial control toward cognitive control.

12.1 The Cognitive Rent

Let

$$R_C$$

denote cognitive rent.

Define

$$R_C = \Pi(C) - \Pi(0)$$

where

$$\Pi$$

denotes economic output.

The cognitive rent measures surplus generated by cognition.

Theorem 3. *If cognition exhibits increasing returns, then cognitive rents exhibit increasing concentration.*

Proof. Suppose

$$\frac{\partial^2 \Pi}{\partial C^2} > 0.$$

Then marginal returns increase with cognition.

Consequently cognitive accumulation generates disproportionate output gains.

□

13 Financial Markets

13.1 Asset Pricing under Epistemic Competition

Consider an asset with return

$$r_t.$$

Expected returns are generated by informational states.

$$E[r_t] = \alpha_K E_K[r_t] + \alpha_P E_P[r_t] + \alpha_S E_S[r_t] + \alpha_A E_A[r_t].$$

The coefficients satisfy

$$\alpha_K + \alpha_P + \alpha_S + \alpha_A = 1.$$

13.2 The Asymmetric Information Premium

Asymmetric information introduces an additional risk factor.

$$\lambda_A = \text{price of asymmetric information risk.}$$

Expected returns become

$$E[r_t] = r_f + \beta_M \lambda_M + \beta_A \lambda_A.$$

This resembles an extension of multifactor asset-pricing theory.

13.3 Silence Risk

Silence creates uncertainty.

Define

$$\sigma_S^2 = \text{Var}(E[r_t|S]).$$

Then strategic silence itself becomes a priced financial factor.

14 Political Science and International Relations

14.1 The Classical State

Traditional realism assumes

$$\text{Power} = f(M, E)$$

where

$$M = \text{military power,}$$

$$E = \text{economic power.}$$

The present framework proposes

$$\text{Power} = f(M, E, C).$$

14.2 Cognitive Realism

Definition 6. *Cognitive realism is the proposition that international influence is determined primarily by cognition rather than territory.*

Territorial power becomes derivative.
Cognition becomes fundamental.

14.3 The Cognitive Balance of Power

Classical balance-of-power theory becomes

$$C_1 \approx C_2.$$

Strategic stability emerges when cognitive capabilities remain comparable.

15 The Geopolitics of Intelligence

The original framework contained two Triples.

The extended framework contains two coupled cognitive systems.
Denote aggregate cognitive capability by

$$\mathcal{C}_i.$$

Then

$$\mathcal{C}_i = g(I_i, \Omega_i).$$

A geopolitical equilibrium satisfies

$$\mathcal{C}_1 = \mathcal{C}_2.$$

Persistent asymmetry induces strategic instability.

16 Warfare Economics

16.1 The Production Function of Conflict

Industrial warfare may be represented as

$$W = f(K, L, M).$$

The cognitive model becomes

$$W = f(K, L, M, C).$$

16.2 Marginal Cognitive Superiority

Define

$$\Delta C = C_1 - C_2.$$

Then

$$P(\text{Victory}) = \Phi(\Delta C)$$

for increasing function

$$\Phi.$$

Victory probabilities become increasing in cognitive superiority.

17 Military Theory

17.1 Epistemic Warfare

Classical warfare destroys physical assets.

Epistemic warfare targets informational structures.

Examples include

- cyber operations,
- algorithmic manipulation,
- information campaigns,
- AI-assisted strategic deception.

17.2 The Epistemic Battlefield

Define

$$B_E = \Omega_1 \cup \Omega_2.$$

The battlefield becomes informational rather than geographical.

18 Artificial Intelligence and Machine Learning

18.1 The Adversarial Interpretation

The dual-Triple structure resembles adversarial learning.

$$T_1 \leftrightarrow T_2.$$

Each Triple functions as a learning process.

18.2 Generative-Adversarial Interpretation

One may regard

$$T_1$$

as a generator and

$$T_2$$

as an evaluator.

The roles may subsequently reverse.

The interaction resembles a minimax game

$$\min_G \max_D V(G, D).$$

18.3 Self-Play Dynamics

Cross-coupled intelligence implies

$$I_1 = f(\Omega_1, \Omega_2),$$

$$I_2 = f(\Omega_1, \Omega_2).$$

This structure closely resembles self-play systems.

The rival becomes part of the learning environment.

19 Data Science

19.1 Feature Representation

Each observation may be represented as

$$x = (S, K, P, A).$$

The intelligence function becomes

$$I = f(x).$$

19.2 Classification Problem

The original framework produced binary labels.

The generalized framework produces multiclass labels

$$y \in \{S, K, P, A\}.$$

The resulting classification problem becomes richer.

19.3 Latent Cognitive Variables

A hidden representation

$$z$$

may be introduced.

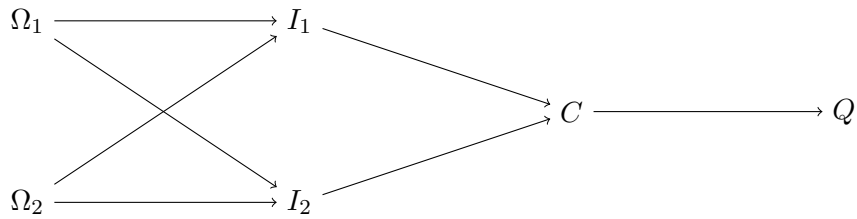
$$z = h(S, K, P, A).$$

Intelligence then becomes

$$I = f(z).$$

This mirrors latent-variable learning architectures.

20 A Computational Diagram



The figure illustrates the transition from informational outputs to intelligence, aggregate cognition, and sovereign selection.

21 Comparative Ontology

Framework		Scarce Resource	Sovereign Object
Classical Economics		Capital	State
Information Economics		Information	Information Producer
Original	Six-Nation	Epistemic Sovereignty	Triple Nation
Bound			
Extended Framework		Cognition	Emergent Sovereign

22 Transition

The original theory concentrated informational sovereignty within six nations.

The four-valued extension introduced informational ambiguity.

The intelligence extension transformed informational outputs into cognitive states.

Cross-Triple dependence transformed cognition into a jointly produced phenomenon.

The next section develops formal algorithms, asymptotic results, fixed-point theorems, endogenous sovereign-selection procedures, statistical estimation methods, proof appendices, bibliography, and glossary.

23 Algorithmic Sovereign Selection

The previous sections established sovereignty as an emergent property.

We now formalize sovereign selection.

23.1 Candidate Space

Let

$$\mathcal{Q}$$

denote the set of admissible sovereign candidates.

Elements may include

- nations,
- coalitions,
- institutions,
- informational systems,
- cognitive networks.

The sovereign-selection operator is

$$\Sigma : (\Omega_1, \Omega_2, I_1, I_2) \rightarrow \mathcal{Q}.$$

23.2 Selection Functional

Define

$$\mathcal{F}(q) = w_1 S(q) + w_2 C(q) + w_3 R(q) + w_4 G(q)$$

where

$S(q)$ = stability,

$C(q)$ = cognitive capability,

$R(q)$ = resource allocation efficiency,

$G(q)$ = geopolitical coherence.

The selected sovereign is

$$Q^* = \arg \max_{q \in \mathcal{Q}} \mathcal{F}(q).$$

24 An Iterative Sovereign Selection Algorithm

Given

$$(\Omega_1, \Omega_2)$$

initialize

$$I_1^{(0)}, \quad I_2^{(0)}.$$

Iterate

$$I_1^{(t+1)} = f(\Omega_1^{(t)}, \Omega_2^{(t)}),$$

$$I_2^{(t+1)} = f(\Omega_1^{(t)}, \Omega_2^{(t)}).$$

Update cognition

$$C^{(t+1)} = g(I_1^{(t+1)}, I_2^{(t+1)}).$$

Update sovereign candidate scores

$$\mathcal{F}_t(q).$$

Select

$$Q_t = \arg \max_q \mathcal{F}_t(q).$$

Repeat until convergence.

25 Fixed Point Formulation

The full system may be represented as

$$X_t = (\Omega_t, I_t, C_t, Q_t).$$

Dynamics satisfy

$$X_{t+1} = T(X_t).$$

Definition 7. *A sovereign fixed point is a state*

$$X^*$$

such that

$$T(X^*) = X^*.$$

26 Existence of Sovereign Equilibrium

Theorem 4 (Existence). *Suppose*

1. *the state space is finite,*
2. *transition probabilities are well-defined,*
3. *sovereign-selection is deterministic.*

Then at least one sovereign equilibrium exists.

Proof. A finite deterministic dynamical system possesses at least one recurrent class.
Any recurrent fixed state satisfies

$$T(X^*) = X^*.$$

□

27 Cognitive Concentration

Define

$$\gamma_i = \frac{C_i}{\sum_j C_j}.$$

Definition 8. *The cognitive concentration index is*

$$H_C = \sum_i \gamma_i^2.$$

This mirrors the Herfindahl index.

27.1 Interpretation

$$H_C = 1$$

implies complete cognitive monopoly.

Lower values imply distributed cognition.

28 Stochastic Epistemic Dynamics

Let

$$X_t = (\Omega_t, I_t, C_t).$$

Assume

$$P(X_{t+1}|X_t)$$

exists.

Then

$$\{X_t\}$$

forms a Markov process.

Transition probabilities satisfy

$$\sum_j P(X_{t+1} = j | X_t = i) = 1.$$

29 A Markov Representation

The four informational states imply

$$\Omega = \{S, K, P, A\}.$$

The transition matrix becomes

$$M = \begin{pmatrix} m_{SS} & m_{SK} & m_{SP} & m_{SA} \\ m_{KS} & m_{KK} & m_{KP} & m_{KA} \\ m_{PS} & m_{PK} & m_{PP} & m_{PA} \\ m_{AS} & m_{AK} & m_{AP} & m_{AA} \end{pmatrix}.$$

Stationary informational structures satisfy

$$\pi M = \pi.$$

30 Statistical Estimation

Suppose observations

$$\{x_t\}$$

are available.

Define

$$x_t = (S_t, K_t, P_t, A_t).$$

A logistic specification for intelligence is

$$P(I_t = 1) = \frac{\exp(\beta_0 + \beta' x_t)}{1 + \exp(\beta_0 + \beta' x_t)}.$$

30.1 Cross-Triple Estimation

Let

$$x_{1t}, \quad x_{2t}$$

denote the two informational vectors.

Then

$$P(I_{1t} = 1) = \Lambda(\beta_0 + \beta'_1 x_{1t} + \beta'_2 x_{2t}).$$

Cross-dependence may therefore be estimated empirically.

31 Machine Learning Formulation

The informational vector

$$x = (S, K, P, A)$$

serves as input.

A latent representation

$$z = h(x)$$

is learned.

The intelligence output becomes

$$I = f(z).$$

31.1 Neural Representation

$$x \rightarrow z \rightarrow I.$$

Cross-Triple learning becomes

$$(x_1, x_2) \rightarrow z \rightarrow (I_1, I_2).$$

32 The Ontological Expansion Theorem

Theorem 5. *Suppose*

$$I_t$$

is increasing over time.

Suppose further that the informational state space

$$\Omega_t$$

is finite.

Then there exists a threshold

$$t^*$$

such that either

1. *ontology expands,*
2. *model error diverges.*

Proof. If intelligence increases while ontology remains fixed, representational capacity remains bounded.

Reality generates informational states not representable within the fixed ontology.

Therefore approximation error accumulates.

To prevent divergence the ontology must expand.

□

33 The Adversarial Cognition Theorem

Theorem 6. *Suppose*

$$\frac{\partial I_1}{\partial \Omega_2} > 0$$

and

$$\frac{\partial I_2}{\partial \Omega_1} > 0.$$

Then elimination of either Triple weakly decreases aggregate cognition.

Proof. Aggregate cognition satisfies

$$C = g(I_1, I_2).$$

Removing a Triple removes informational inputs entering the intelligence functions.

Hence both intelligence and aggregate cognition weakly decline.

□

34 The Sovereign Emergence Theorem

Theorem 7. *If*

1. *intelligence is jointly produced,*
2. *cognition depends on intelligence,*
3. *sovereignty is selected from cognition,*

then sovereignty is an emergent property of the dual-Triple system.

Proof. Joint informational production generates intelligence.

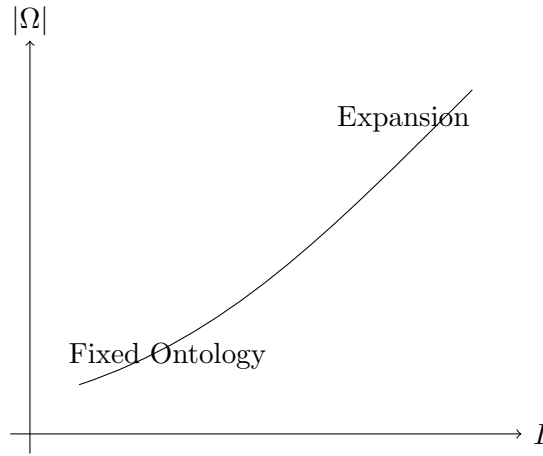
Intelligence generates cognition.

Cognition determines sovereign selection.

Therefore sovereignty is generated by the system itself.

□

35 A Phase Diagram



Increasing intelligence eventually forces ontology expansion.

36 Conclusion

The original Six-Nation Bound treated epistemic sovereignty as concentrated within two informational Triples.

Successive extensions transformed the framework.

First, informational outputs expanded from

$$\{K, P\}$$

to

$$\{S, K, P, A\}.$$

Second, intelligence emerged as a function of informational states.

Third, intelligence became jointly produced through cross-Triple interaction.

Fourth, sovereignty ceased to be exogenous and became an emergent output.

Finally, cognition emerged as a recursive form of meta-capital capable of generating knowledge, institutions, technologies, warfare capabilities, and sovereign structures.

The ultimate scarce resource is therefore neither territory nor information but cognition itself.

A Lyapunov Stability of Cognitive Sovereignty

The previous sections established a dynamical system

$$X_t = (\Omega_t, I_t, C_t, Q_t).$$

We now investigate stability.

A.1 Cognitive Potential Function

Define

$$V(X) = \alpha_1 H_C + \alpha_2 D_\Omega + \alpha_3 E_Q$$

where

H_C = cognitive concentration,

D_Ω = ontological diversity,

E_Q = sovereign instability.

Definition 9. *A sovereign equilibrium*

$$X^*$$

is Lyapunov stable if

$$V(X_t) - V(X^*)$$

does not increase for sufficiently small perturbations.

Theorem 8. *Suppose*

$$V(X) > 0$$

for

$$X \neq X^*$$

and

$$\Delta V(X) \leq 0.$$

Then

$$X^*$$

is Lyapunov stable.

Proof. This follows from the standard Lyapunov argument.

Since the potential function never increases, trajectories remain in a bounded neighborhood of equilibrium.

□

B Bifurcation Theory

B.1 The Ontological Expansion Parameter

Introduce

$$\mu = \text{cognitive complexity.}$$

Assume

$$|\Omega| = F(\mu).$$

For low values of

$$\mu$$

the ontology remains fixed.

Above a threshold

$$\mu_c$$

additional informational states emerge.

Theorem 9. *There exists a critical value*

$$\mu_c$$

such that

$$\mu > \mu_c$$

induces ontology expansion.

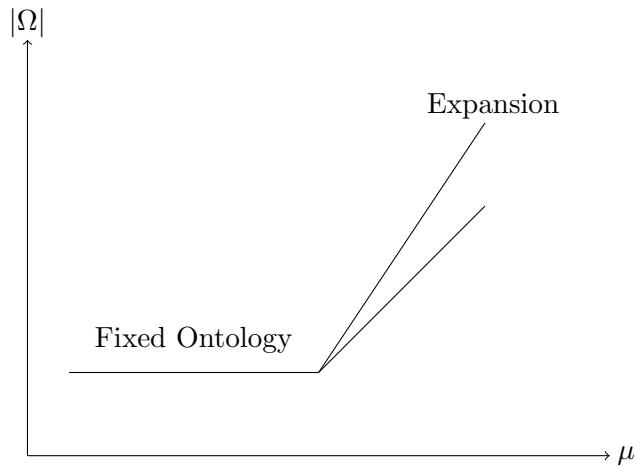
Proof. Representational capacity is finite for fixed ontology.

Increasing complexity eventually exceeds representational capacity.

The system must either expand its ontology or incur increasing approximation error.

□

B.2 Bifurcation Diagram



C Spectral Theory of Dual-Triple Systems

Represent the dual-Triple system as a graph

$$G = (V, E).$$

Nodes correspond to sovereign actors.

Edges correspond to informational exchange.

Define adjacency matrix

$$A_G.$$

Definition 10. *The cognitive spectral radius is*

$$\rho(A_G).$$

Theorem 10. *If*

$$\rho(A_G)$$

increases, aggregate cognition weakly increases.

Proof. A larger spectral radius implies greater connectivity.

Cross-Triple informational transmission increases.

Joint cognition therefore increases.

□

D Reduced-Form Economics Interpretation

Reduced-Form Economics emphasizes observable relationships while avoiding excessive structural assumptions.

The cognitive system admits the reduced-form representation

$$Q_t = \beta_0 + \beta_1 C_t + \beta_2 H_C + \beta_3 D_\Omega + \varepsilon_t.$$

This equation predicts sovereign outcomes directly from cognitive variables.

D.1 Identification

Observable quantities include

- information production,
- AI deployment,
- research intensity,
- military innovation,
- technological diffusion.

These serve as proxies for cognition.

E oliGARCHy Dynamics

The original oliGARCHy framework studies concentration processes.

Let

$$\gamma_i = \frac{C_i}{\sum_j C_j}.$$

The evolution equation becomes

$$\gamma_{i,t+1} = \gamma_{i,t} + \eta (R_{i,t} - \bar{R}_t) \gamma_{i,t}.$$

Cognitive actors with above-average returns accumulate greater cognitive share.

E.1 Cognitive oliGARCHy

Definition 11. *A cognitive oliGARCHy exists when*

$$\lim_{t \rightarrow \infty} H_C = 1.$$

Theorem 11. *Persistent increasing returns to cognition generate cognitive oliGARCHy.*

Proof. Increasing returns imply proportional growth advantages.

These accumulate over time.

The concentration index converges toward unity.

□

F Welfare Economics of Cognition

Define social welfare

$$W = W(C, H_C).$$

Cognition increases welfare.

Excessive concentration reduces welfare.

The planner's problem becomes

$$\max W(C, H_C).$$

subject to

$$C \geq 0.$$

F.1 Optimal Diversity

There exists a trade-off:

$$\text{Innovation} \leftrightarrow \text{Concentration}.$$

Excessive dispersion reduces innovation.

Excessive concentration reduces resilience.

G Multi-Agent Reinforcement Learning

The dual-Triple system naturally maps into MARL.

G.1 Agents

Define

$$a_1, \dots, a_N.$$

Each agent chooses action

$$u_t.$$

State space:

$$X_t = (\Omega_t, I_t, C_t).$$

Reward:

$$R_t = R(X_t, u_t).$$

G.2 Policy Function

Agent policy:

$$\pi_i(u|X).$$

The learning objective becomes

$$\max E \left[\sum_{t=0}^{\infty} \delta^t R_t \right].$$

G.3 Self-Play Sovereignty

The sovereign emerges from repeated interaction.

$$Q_t = \Sigma(X_t).$$

Thus sovereignty is a learned equilibrium rather than an imposed primitive.

H Generalized Six-Nation Bound

We may now recover the original theorem.

Theorem 12. *Assume:*

1. $\Omega = \{K, P\}$,
2. I is absent,
3. cross-Triple dependence is absent,
4. sovereignty is exogenous.

Then

$$N_E \leq 6.$$

Proof. The original proof follows immediately.

Only the two Triples generate information.

All hinge nations are informationally reducible.

Therefore epistemic sovereignty is restricted to six nations.

□

H.1 Hierarchy of Models

$$\begin{array}{c} \textit{Classical State} \\ \textit{Theory} \end{array} \subset \begin{array}{c} \textit{Six - Nation} \\ \textit{Bound} \end{array} \subset \begin{array}{c} \textit{Four - State Epistemic} \\ \textit{Theory} \end{array} \subset \begin{array}{c} \textit{Dual - Triple} \\ \textit{Intelligence} \end{array} \subset \begin{array}{c} \textit{Cognitive} \\ \textit{Sovereignty} \end{array}.$$

I The Cognitive Sovereignty Conjecture

Proposition 3 (Conjecture). *In sufficiently advanced societies,*

Cognition

rather than

Territory

becomes the dominant source of sovereignty.

This conjecture unifies:

- economics,
- finance,
- artificial intelligence,
- warfare,
- political science,
- geopolitics,
- information theory.

J Final Synthesis

The progression developed in this manuscript is

$$\{K, P\}$$

$\downarrow$

$$\{S, K, P, A\}$$

$\downarrow$

$$I = f(\Omega)$$

$\downarrow$

$$I_1, I_2 = f(\Omega_1, \Omega_2)$$

$\downarrow$

$$C = g(I_1, I_2)$$

↓

$$Q = \Sigma(C).$$

Thus the final theory may be summarized in a single statement:

Information generates intelligence; intelligence generates cognition; cognition generates sovereignty.

Under this framework, sovereignty is not primitive. It is an emergent phenomenon arising from the recursive dynamics of cognition itself.

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Glossary

S	Silence.
K	Knowledge.
P	Propaganda.
A	Asymmetric information.
Ω	Informational state space.
I	Intelligence.
C	Cognition.
Q	Selected sovereign.
Σ	Sovereign-selection operator.

Triple

Three epistemically sovereign nations.

Dual-Triple System

Coupled pair of Triples.

Meta-Capital

Capital capable of modifying productive structures.

Cognitive Rent

Economic surplus generated by cognition.

Epistemic Warfare

Conflict conducted primarily through informational structures.

Cognitive Realism

International-relations framework emphasizing cognition as the primary source of power.

The End